InteractAgent: Agentic Human Motion Interaction with Memory-Augmented LLMs

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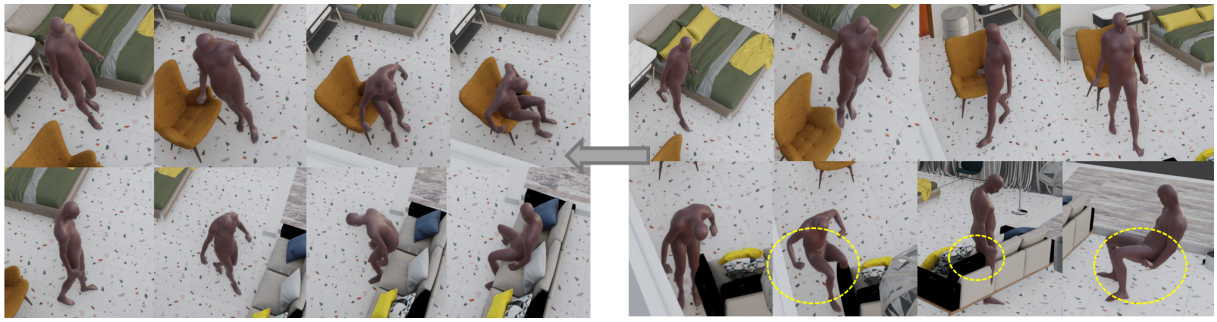

Figure 1: **InteractAgent vs. Naive:** *InteractAgent* (left) couples Online Observation with Memory-Augmented Reflection, iteratively refining motion plans to achieve long-horizon, physically plausible human–scene interactions from images and language only. A naive text-to-motion baseline (right) lacks feedback and quickly accumulates errors, resulting in severe penetration and incomplete tasks.

Abstract

Generating long-horizon human-scene interactions requires coordinated spatial reasoning, physical feasibility, and the ability to adapt actions based on visual feedback. Existing text-driven motion models operate in scene-agnostic latent spaces and ignore environmental constraints, while geometry-conditioned approaches rely on paired scene-motion training and fail to generalize to novel layouts or open-vocabulary tasks. As a result, current systems lack closed-loop correction and accumulate errors over long horizons. We introduce **InteractAgent**, an LLM-agent, reflection-driven framework that generates scene-aware human motion directly from images and natural-language goals. The system integrates two complementary mechanisms: (1) **Online Observation**, which performs step-wise MLLM evaluation to decide whether to continue, redo, or replan each atomic action, preventing error propagation; and (2) **Memory-Augmented Reflection**, which aggregates planning-execution trajectories across attempts and performs multi-round refinement with an adaptive stopping strategy. Without any geometry-conditioned training or 3D preprocessing, InteractAgent boosts long-horizon success over training-free single-pass baselines (up to +59%) and, despite being training- and scene-modeling-free, exceeds the geometry-trained UniHSI model in success (+25.9%) at the cost of higher penetration. Code: `https://github.com/AIGeeksGroup/InteractAgent`.

1 Introduction

Understanding and synthesizing human behavior within complex environments is fundamental to embodied artificial intelligence. A key challenge lies in generating *long-horizon human-scene interactions*: temporally extended motion sequences where agents navigate, manipulate objects, and accomplish tasks in structured 3D spaces. Such scenarios require joint reasoning over spatial layouts, physical constraints, and semantic goals while maintaining temporal coherence. Recent advances in human motion generation have achieved impressive progress through large-scale pretraining (Guo et al., 2022; Jiang et al., 2023; Zhang et al., 2023). However, these models typically operate in *scene-agnostic* latent spaces—motion is generated independently of environmental context, lacking awareness of object locations, affordances, or contact constraints. A generated "sit on the chair" motion

may exhibit plausible kinematics in isolation, yet produce severe penetration when placed in a real scene.

However, generating scene-aware, long-horizon human–scene interactions remains challenging. Existing methods fall into two paradigms with inherent limitations. Geometry-conditioned models explicitly enforce mesh-level constraints or contact priors (Hassan et al., 2021b; Xiao et al., 2024; Jiang et al., 2024), but rely on paired motion–scene annotations, accurate 3D geometry, and domain-specific supervision, which limits scalability to novel layouts and open-vocabulary tasks. In contrast, training-free text-to-motion models leverage large-scale motion priors but lack geometric grounding: operating in scene-agnostic latent spaces, they cannot reason about spatial relations or physical feasibility. As a result, early deviations cannot be corrected, causing errors to accumulate into penetration, drift, or task failure. Crucially, neither paradigm supports *interactive*, *closed-loop*, or *reflective* planning, which is essential for coherent long-horizon embodied behavior.

These observations motivate a different perspective: *can we achieve scene-aware human motion generation purely through multimodal reasoning and reflection, without any task-specific training or geometry preprocessing.* We hypothesize that explicit reflection—observing outcomes, identifying errors, and iteratively refining plans—provides a powerful inductive bias for long-horizon interactions. This leads to **InteractAgent**, a training-free framework that couples a pretrained multimodal large language model (MLLM) with motion language models (MotionLLMs) to perform zero-shot, scene-grounded motion planning. The system decomposes human–scene interaction into two complementary reasoning loops: (1) **Online Observation**, a step-wise feedback mechanism that evaluates each executed motion and decides whether to continue, redo, or replan; and (2) **Memory-Augmented Reflection**, a multi-round task-level improvement process that aggregates planning–execution history with an adaptive stopping criterion. Together, these mechanisms connect *local closed-loop robustness* with *global long-horizon optimization*, enabling self-correcting and interpretable interaction sequences without finetuning.

The contributions of this paper can be summarized as follows:

- We propose **InteractAgent**, a training-free reflective framework for long-horizon human-scene interaction, integrating scene understanding, task decomposition, online feedback, and motion synthesis via multimodal prompting, without geometry-conditioned training or task-specific finetuning.
- We introduce an *online observation* and *memory-augmented reflection* mechanism for closed-loop interaction, in which long-horizon interaction history is represented as a single trajectory visualization image. This trajectory-level visual representation serves as a compact reflection signal for the LLM, enabling step-wise correction, multi-round refinement, and adaptive termination without explicit geometric state supervision.
- Extensive experiments on UniHSI and TRUMANS which improves success rates from 25.5% to **84.3%** and reduces penetration from 528.4 to **48.6** in pe_{max}, significantly reduced penetration errors, and improved interpretability over training-free and single-pass baselines.

2 Related Work

Human motion generation. Human motion generation aims to synthesize realistic, temporally coherent, and semantically meaningful motion under text, audio, or control signals. Early work relied on recurrent or transformer-based models trained on MoCap datasets. Text-to-motion methods such as Guo et al. (Guo et al., 2022) and T2M-GPT (Zhang et al., 2023) learn a shared latent space or VQ-VAE codebook to map text to discrete motion tokens, while MotionGPT (Jiang et al., 2023) unifies motion and text as a single "language". MoMask (Guo et al., 2024) improves temporal coherence via hierarchical quantization and masked motion modeling. Diffusion-based approaches further enhance realism and controllability: Pi et al. (Pi et al., 2023) use hierarchical diffusion for detailed human-object motion, and Wang et al. (Wang et al., 2025) propose a cross-modal encoder-decoder mixture for text-driven 3D motion. However, these models operate in scene-agnostic latent spaces and do not explicitly account for environmental constraints, contacts, or navigation, limiting their use in embodied, scene-grounded settings.

Human motion interaction. To go beyond single-agent motion, prior work studies human-human (HHI), human–object (HOI), and human-scene (HSI) interactions. HOI frameworks (Li et al., 2023a; Taheri et al., 2020; Li et al., 2024) model affordances and contact dynamics through

object-conditioned synthesis, grasping datasets, or controllable interaction goals. HSI methods (Hassan et al., 2021b) incorporate 3D layouts and pose–scene embeddings to generate physically plausible motion in structured environments. A variety of datasets and simulators support grounded interaction learning: BEHAVE (Bhatnagar et al., 2022) captures multi-view RGB-D human-object interactions, HOISim (Zakour et al., 2021) provides simulation-based HOI data, and Wang et al. (Wang et al., 2021) generate long-term motion-in-scene sequences. Holistic++ (Chen et al., 2019) jointly infers layout, human pose, and affordances, while Wu et al. (Wu et al., 2025b) map natural-language instructions to 3D interactive motions. These approaches significantly improve realism but rely on hand-crafted scene features or geometry-conditioned training, and they lack the high-level semantic reasoning needed for adaptive planning in novel environments.

Vision-language models for scene understanding. Vision-language models (VLMs) and multimodal LLMs (MLLMs) enable cross-modal reasoning and grounding between perception and language. BLIP-2 (Li et al., 2023b) and PaLM-E (Driess et al., 2023) align visual encoders with large language models for semantic localization, reasoning, and action grounding. VIMA (Jiang et al., 2022) and RT-2 (Zitkovich et al., 2023) further demonstrate that multimodal models can serve as general-purpose planners for embodied control. Qwen-VL (Bai et al., 2023) improves high-resolution visual reasoning and spatial understanding. 3D-aware extensions such as Scene-LLM (Fu et al., 2024) and 3D-LLM (Hong et al., 2023) inject 3D geometry for 3D-aware reasoning and manipulation. Chen et al. (Chen et al., 2025) use VLMs to mine pose knowledge for human pose estimation, while EgoNav (Wang et al., 2024), UniEgoMotion (Patel et al., 2025), and Wang et al. (Wang et al., 2023) tackle egocentric motion forecasting and pose estimation by leveraging human-scene relationships. Human-centric metrics (Fu et al., 2020) quantify spatial affordance and reachability. Together, these works suggest that coupling a grounding-capable MLLM with tokenized motion models (e.g., MotionGPT, T2M-GPT, MoMask) can support a perception-planning-generation pipeline. Our framework builds on this idea and explicitly connects MLLM-based scene analysis, reflective planning, and a MotionLLM executor to achieve end-to-end motion generation from visual perception to 3D realization.

3 Method

3.1 Overview

InteractAgent generates long-horizon, scene-aware human motion entirely through training-free multimodal reasoning. Given one or more scene images and a high-level instruction, the system operates in three stages: (1) multimodal scene analysis, (2) action-level plan generation, and (3) motion execution using a frozen MotionLLM controller. Crucially, InteractAgent augments this pipeline with **dual-layer reflection**: step-wise *Online Observation* for local correction and *Memory-Augmented Reflection* for global trajectory refinement.

A single multimodal LLM (GPT-4o in all main experiments) performs all high-level reasoning: it parses the scene, decomposes the task into atomic actions, judges execution outcomes, and updates future plans based on accumulated evidence. The MotionLLM module Q (from the Motion-Agent framework) is used strictly as a low-level executor that converts natural-language actions into 3D motion segments. We treat Q as a black-box controller: no finetuning or geometry-specific adaptation is performed.

3.2 Scene Motion Planning Pipeline

InteractAgent establishes an end-to-end pipeline that connects visual perception with motion synthesis. Given a scene observation $\mathbf{s}$ (e.g., RGB images or multi-view renderings) and a high-level task description $\mathbf{t}$, the planner produces a structured motion sequence $\mathbf{x} = \{\mathbf{x}_1, \ldots, \mathbf{x}_L\}$ that realizes the intended behavior within the environment. An overview of the full pipeline is shown in Fig. 2. In particular, we encode the long-horizon interaction history as a single trajectory visualization image (Fig. 3) and use it as a compact reflection signal for the MLLM.

Formally, we denote the underlying motion executor by Q, which maps a sequence of natural-language action descriptions to a full motion trajectory:

$$Q: \ \mathbf{p} = \{\mathbf{p}_1, \ldots, \mathbf{p}_K\} \ \longrightarrow \ \mathbf{x}.$$

For clarity, we omit the implicit dependence of Q on the scene $\mathbf{s}$ and task $\mathbf{t}$, and refer to it simply as the MotionLLM executor used throughout Algorithms 1 and 2.

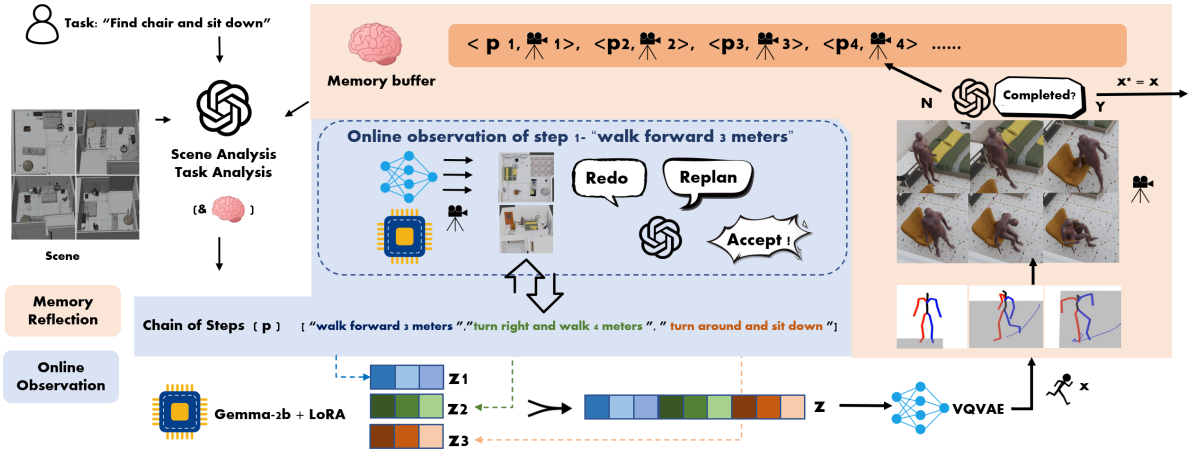

Figure 2: **Overview of InteractAgent.** Our framework couples a single multimodal LLM with a frozen MotionLLM executor. Two key components enable training-free, long-horizon interaction generation: (1) *Online Observation* provides step-wise visual feedback during execution, allowing the MLLM to correct local deviations; (2) *Memory-Augmented Reflection* aggregates planning–execution histories across rounds and refines future plans at the trajectory level. Together, these mechanisms form a dual-layer reflective loop that progressively improves motion quality and scene consistency.

Figure 3: Trajectory visualization image used as a compact reflection signal for the MLLM.

Scene Understanding. To initialize the process, the scene is first provided as a prompt to the MLLM, which interprets spatial layouts, object affordances, and potential interaction sites:

$$\mathbf{g} = f_{\text{MLLM}}(\mathbf{s}).$$

The scene representation $\mathbf{g}$ encodes entities, relations, and spatial structure for downstream reasoning.

Action Sequence Generation. Conditioned on $\mathbf{g}$ and the task prompt $\mathbf{t}$, the same MLLM decomposes the task into a sequence of atomic, executable natural-language actions:

$$\mathbf{p} = \{\mathbf{p}_1, \ldots, \mathbf{p}_K\}, \quad \mathbf{p}_k \in \text{NL}.$$

Each action describes a short, unambiguous motion primitive (e.g., "turn left", "walk forward 1.5 meters").

Action-to-Motion Execution. We do not design a new motion synthesis model. Instead, InteractAgent directly calls the frozen MotionLLM executor to realize the entire action list:

$$\mathbf{x} = Q(\mathbf{p}).$$

Internally, MotionLLM encodes the natural-language sequence, predicts motion tokens, decodes them with a VQ-VAE motion decoder, and reconstructs SMPL-X parameters. We treat this process as an internal implementation detail of Q and focus on how to generate and refine $\mathbf{p}$ via reflection.

This pipeline constitutes the forward execution path of InteractAgent-from perceptual grounding to motion realization on top of a fixed MotionLLM executor. To enhance robustness through step-wise feedback, we introduce Online Observation (Section 3.3) and Memory-Augmented Reflection (Section 3.4), both implemented via the same MLLM backbone.

3.3 Online Observation

While the scene motion planning pipeline provides a structured perception-to-action flow, it executes the entire action sequence in an open-loop manner, assuming all predicted actions are correct once generated. In practice, early-step inaccuracies (e.g., over- or under-estimated displacements) often prop-

agate through subsequent actions, resulting in physically inconsistent or semantically implausible motions. Such forward error accumulation not only degrades overall realism but also complicates global reflection, as it becomes difficult to isolate whether the degradation stems from local planning errors or model stochasticity.

To alleviate this issue, we introduce an **Online Observation** mechanism that integrates step-wise reflection and local replanning during execution. After each atomic action $\mathbf{p}_k$ is generated and executed by the MotionLLM executor,

$$\mathbf{x}_k = Q(\mathbf{p}_k),$$

the resulting visual outcome $\mathbf{v}_k$ is rendered (e.g., as a short clip or keyframes) and evaluated through the same MLLM, forming an observation pair $(\mathbf{p}_k, \mathbf{v}_k)$.

The MLLM evaluates the alignment between planned action $\mathbf{p}_k$ and rendered outcome $\mathbf{v}_k$, producing a score and textual feedback:

$$(\text{score}_k, \text{feedback}_k) = f_{\text{MLLM}}(\mathbf{p}_k, \mathbf{v}_k).$$

Based on this feedback, the language interface of the MLLM makes a discrete decision

$$d_k \in \{\text{continue}, \text{redo}, \text{replan}\},$$

corresponding to one of three strategies:

- **Continue:** if $\text{score}_k > \tau_{\text{accept}}$, accept the current action $\mathbf{p}_k$ and proceed with the next planned step;
- **Redo:** detect misalignment in the current action, locally correct and regenerate $\mathbf{p}_k$ before continuing the sequence;
- **Replan:** detect trajectory drift, accept $\mathbf{p}_k$ but regenerate the remaining actions $\{\mathbf{p}_{k+1}, \ldots, \mathbf{p}_K\}$ conditioned on the updated scene state.

Through this process, each step produces an optimized action $\mathbf{p}_k^*$, and the planner maintains an updated list of refined actions

$$\mathbf{p}^* = \{\mathbf{p}_1^*, \mathbf{p}_2^*, \ldots, \mathbf{p}_K^*\}.$$

This list captures the locally adjusted and globally consistent plan emerging from the online reflection loop. After completing all steps, the optimized sequence is executed once more by the MotionLLM executor to obtain the final trajectory:

$$\mathbf{x}^* = Q(\mathbf{p}^*).$$

The full procedure is summarized in Algorithm 1.

Algorithm 1 Online Observation (Step-wise)

Input: Scene $\mathbf{s}$; initial task plan $\mathbf{p} = \{\mathbf{p}_1, \ldots, \mathbf{p}_K\}$; MotionLLM executor Q; MLLM evaluator f_{MLLM}; max retries per step $R_{\max}$.

Output: Refined action list $\mathbf{p}^*$, final motion $\mathbf{x}^*$

```
p* ← []
for k = 1 to K do
    retry_count ← 0  ▷ Initialize retry counter
    repeat
        x_k ← Q(p_k)
        v_k ← Render(x_k)
        (score_k, feedback_k) ← f_MLLM(p_k, v_k)
        d_k ∈ {continue, redo, replan} based on feedback_k
        if d_k ≠ continue then
            retry_count ← retry_count + 1
        end if
        if d_k = redo then
            regenerate p_k using MLLM feedback
        else if d_k = replan then
            replan {p_{k+1}, ..., p_K} conditioned on current state
        end if
    until d_k = continue or retry_count ≥ R_max
    p*_k ← p_k  ▷ Accept last attempt
end for
x* ← Q(p*)
return p*, x*
```

By embedding reflection within each execution step, InteractAgent continuously aligns its planned actions with the observed outcomes, forming a closed-loop perception-action cycle. This design transforms the planner from an offline generator into an *execution-aware interactive system* that perceives, acts, and self-adjusts in real time. To prevent excessive local oscillation, Online Observation enforces a maximum number of correction attempts for each action step. If repeated *redo* or *replan* decisions exceed this limit, the system terminates early and accepts the most recent action $\mathbf{p}_k$ as the final outcome for that step. Once the limit is reached, the planner does not attempt additional adjustments and proceeds using only the last generated version of the action, discarding previous failed attempts.

While Online Observation provides stable step-wise correction, it remains limited to short-horizon

improvements. To further enhance long-term consistency and task-level reasoning, we introduce a **Memory-Augmented Reflection** mechanism.

3.4 Memory-Augmented Reflection

Online Observation focuses on local corrections within a single execution of a task. However, long-horizon behaviors often require global adjustments that can only be discovered by comparing *multiple* executions of the same task. To this end, we augment InteractAgent with a trajectory-level memory that accumulates planning-execution pairs across attempts and enables global reflection.

Iterative Reflection Process. Given a specific task defined by scene $\mathbf{s}$ and goal $\mathbf{t}$, the model executes the complete planning and motion generation process multiple times. Each execution produces a planned action sequence $\mathbf{p}_i$ and the corresponding rendered outcome $\mathbf{v}_i$, which form a planning–result pair $(\mathbf{p}_i, \mathbf{v}_i)$. These pairs are accumulated as memory:

$$\mathcal{M}_i = \{(\mathbf{p}_1, \mathbf{v}_1), (\mathbf{p}_2, \mathbf{v}_2), \ldots, (\mathbf{p}_i, \mathbf{v}_i)\}.$$

At the $(i+1)$-th iteration, the model performs reflection by reasoning over the accumulated memory $\mathcal{M}_i$ when generating a new plan:

$$\mathbf{p}_{i+1} = f_{\text{reflect}}(\mathcal{M}_i, \mathbf{s}, \mathbf{t}),$$

where f_{reflect} is implemented by prompting the same MLLM f_{MLLM} with the task description, scene context, and all previous planning–execution pairs. The MLLM can directly observe rendered videos $\{\mathbf{v}_1, \ldots, \mathbf{v}_i\}$ to identify patterns of failure—such as persistent penetration errors, trajectory drift, or incomplete task execution—and adjust its planning strategy accordingly.

The newly generated plan $\mathbf{p}_{i+1}$ is then executed and rendered as $\mathbf{v}_{i+1}$, after which the pair is appended to memory:

$$\mathcal{M}_{i+1} = \mathcal{M}_i \cup \{(\mathbf{p}_{i+1}, \mathbf{v}_{i+1})\}.$$

Adaptive Stopping Criterion. Unlike fixed-iteration approaches, our framework incorporates an adaptive stopping mechanism to balance quality and computational efficiency. After each reflection round, the MLLM evaluates the generated motion $\mathbf{v}_i$ against the task requirements $\mathbf{t}$ and provides a judgment on whether the result is satisfactory. If the evaluation confirms that task requirements are met, the system terminates early and outputs the current result. Otherwise, it continues refining until reaching the maximum reflection budget $N_{\max}$ (typically 5 rounds in our experiments).

Algorithm 2 Memory-Augmented Reflection (with Adaptive Stopping)

Input: Scene $\mathbf{s}$; task description $\mathbf{t}$; initial plan $\mathbf{p}_0$; MotionLLM executor Q; MLLM-based evaluator f_{MLLM}; max rounds $N_{\max}$.
Output: Trajectory memory $\mathcal{M}_{\text{final}}$ containing all attempts

$\mathbf{x}_0 \leftarrow Q(\mathbf{p}_0)$
$\mathbf{v}_0 \leftarrow \text{Render}(\mathbf{x}_0)$
$\mathcal{M}_0 \leftarrow \{(\mathbf{p}_0, \mathbf{v}_0)\}$
for $i = 1$ to $N_{\max}$ **do**
  $\mathbf{p}_i \leftarrow f_{\text{reflect}}(\mathcal{M}_{i-1}, \mathbf{s}, \mathbf{t})$
  $\mathbf{x}_i \leftarrow Q(\mathbf{p}_i)$
  $\mathbf{v}_i \leftarrow \text{Render}(\mathbf{x}_i)$
  $\mathcal{M}_i \leftarrow \mathcal{M}_{i-1} \cup \{(\mathbf{p}_i, \mathbf{v}_i)\}$
  $\text{status}_i \leftarrow f_{\text{MLLM}}(\mathbf{p}_i, \mathbf{v}_i, \mathbf{t})$
  if $\text{status}_i = \text{Pass}$ **then**
    break ▷ Task requirements satisfied
  end if
end for
$\mathcal{M}_{\text{final}} \leftarrow \mathcal{M}_i$
return $\mathcal{M}_{\text{final}}$

Final Output. After convergence or reaching $N_{\max}$ rounds, the system outputs the motion from the final iteration:

$$\mathbf{x}_{\text{final}} = Q(\mathbf{p}_i),$$

where i denotes the stopping iteration and Q is the MotionLLM executor defined above. This final motion represents the outcome of a complete self-reflection cycle.

The overall reflection process is summarized in Algorithm 2.

By integrating memory-augmented reflection with adaptive stopping, InteractAgent transforms from a one-shot generator into an *self-improving system* that learns from its own experiences—bridging short-term step-wise correction (Section 3.3) and long-term task-level optimization for robust, scene-aware human motion generation.

4 Experiments

We evaluate our training-free reflection framework on language-driven, long-horizon human-scene interaction tasks. Based on the structure of the framework and the nature of our tasks, we select representative datasets and metrics to assess both task-level

success and physical plausibility. All experiments are designed to verify how reflection improves the quality and consistency of long-horizon human–scene interactions.

4.1 Datasets and Metrics

Dataset Previous human–scene interaction datasets mainly focus on short and constrained interactions (Hassan et al., 2021a; Peng et al., 2021; Zhao et al., 2022), often limited to single-object contact or short motion clips without long-horizon reasoning. To comprehensively evaluate our training-free reflection framework, we draw from two complementary resources.

TRUMANS (Jiang et al., 2024) provides physically detailed indoor scenes with dynamic object configurations, supporting realistic assessment of motion plausibility.

UniHSI (Xiao et al., 2024) introduces language-driven, arbitrary-length interaction tasks, aligning closely with our text-conditioned planning structure.

Metrics Following Hassan et al. (Hassan et al., 2023), we use Success Rate (SR) to measure whether each interaction task is successfully completed. For physical plausibility, we adopt Physical Validity (PV) based on body–scene penetration and report two robust statistics: pe_{max} (maximum penetration) and $\text{pe}_{10\%}$ (top-10% mean over highest-penetration frames). See Appendix A for the full formulation and motivation.

4.2 Implementation Details

Unless otherwise specified, we instantiate the MLLM with GPT-4o via its official API. We adopt the public MotionLLM executor from MotionAgent and keep all released hyperparameters unchanged. We set $N_{\max} = 5$ reflection rounds and allow up to $R_{\max} = 2$ retries per atomic action. Experiments run on a single NVIDIA A6000 GPU.

4.3 Main Result

We evaluate InteractAgent on long-horizon human-scene interaction tasks of varying complexity levels. Table 1 summarizes the task success rate (SR) and physical plausibility metrics across all methods. Our full system (Ref + Online) achieves the strongest performance among all training-free variants and substantially closes the gap to the supervised oracle UniHSI-CoCo.

Methods	**Success Rate (%)** ↑			$\textbf{pe}_{\textbf{max}}$ ↓	$\textbf{pe}_{10\%}$ ↓
	Low	Mid	High		
UniHSI-CoCo	**85.1**	47.7	17.7	**18.2**	**9.3**
Naive	25.5	18.2	4.4	528.4	113.6
Online	74.4	52.3	31.1	102.7	54.5
Ref + Offline	72.3	47.7	26.7	62.3	36.8
InteractAgent	**84.3**	**79.5**	**64.4**	**48.6**	**29.7**

Table 1: **Result of methods on tasks of different complexity levels.** Success Rate (SR, %) and penetration errors (pe_{max}, $\text{pe}_{10\%}$) are reported.

Overall, the full framework demonstrates strong performance across all difficulty levels. In particular, on mid- and high-complexity tasks, InteractAgent (79.5%, 64.4%) significantly surpasses the supervised UniHSI-CoCo planner (47.7%, 17.7%), despite requiring no geometry-conditioned training. This result suggests that reflection driven, training-free reasoning can outperform specialized supervised models when tasks require multi-step spatial reasoning rather than single-clip contact modeling.

In terms of physical plausibility, our method reduces penetration errors by more than an order of magnitude compared with the Naive baseline, achieving pe_{max} = 48.6 and $\text{pe}_{10\%}$ = 29.7. These improvements highlight the effectiveness of our reasoning-driven framework in generating motions that not only complete tasks but also adhere to scene constraints.

The following ablation studies Section 4.4 further decompose the contributions of Online Observation, Memory-Augmented Reflection, and different LLM planners.

4.4 Ablation Study

To better understand each component's contribution, we conduct ablations along three dimensions: (1) the reflection mechanism (Online Observation vs. Memory-Augmented Reflection), (2) the multimodal LLM planner, and (3) the motion executor. Unless otherwise specified, all other settings are kept identical and we vary one factor at a time.

Ablation on Reflection Components. Table 1 already decomposes our system into four variants-Naive, Online Observation only, Offline Memory Reflection, and the full Online Memory Reflection.

- **Online Observation (local feedback).** Enables step-wise correction and prevents early errors from propagating, substantially improving long-horizon task success, leading to large penetration reduction.

LLM Type	SR (%) ↑	PC (%) ↑
Human	72	–
w. Qwen-VL-Max p	56	75
w. GPT-4o	64	81

Table 2: Comparison of SR and PC (Y/N decided by human) across different LLM types.

Motion Executor	SR (%)↑	pe_{max}↓	$pe_{10\%}$↓
MotionLLM	70	48.6	29.7
MG-MotionLLM	72	23.3	16.9
MG stateless	70	46.2	24.3

Table 3: Comparison of SR and penetration between Motion Executors.

- **Memory-Augmented Reflection (global feedback).** Aggregates planning-execution histories across attempts and corrects systematic failure patterns, Optimize the path selection for long tasks and significantly reduce the number of steps.
- **Online vs. Offline Memory.** Trajectory-level memory must be paired with updated scene grounding; otherwise the planner can drift and accumulate errors over long horizons.

Overall, these ablations reveal a clear division of labor: *Online Observation* stabilizes long-horizon execution by correcting early deviations and preventing error accumulation, while enforcing local physical consistency; *Memory-Augmented Reflection* guides the global planning strategy. Their combination (REF + ONLINE) delivers the best performance, and we adopt it as our final model.

Ablation on LLM Planner. We further study how the choice of LLM affects planning robustness. Table 2 compares GPT-4o and Qwen-VL-Max under identical settings. GPT-4o excels at long-horizon reasoning and yields higher Planning Correctness (81% vs. 75%) and SR (64% vs. 56%), while Qwen-VL-Max provides competitive visual grounding but struggles with complex multi-step reasoning.

Ablation on MotionLLM Executor While InteractAgent is designed to be executor-agnostic, the quality of the low-level motion generator can still influence the final physical validity of long-horizon trajectories. To isolate this factor, we conduct an ablation study by replacing the default MotionLLM executor Q with alternative motion backbones, while keeping *all* high-level components unchanged. The results are summarized in Table 3.

Although MG-MotionLLM (Wu et al., 2025a) does not improve task success rate compared with MotionLLM, it significantly reduces penetration. This suggests that task completion is mainly constrained by high-level planning and scene reasoning, while physical plausibility depends on the motion executor.

Summary. Our ablation results demonstrate that (1) Online Observation and Memory-Augmented Reflection address complementary failure modes, together enabling robust long-horizon interaction; (2) the LLM planner strongly influences reasoning quality but does not require any task-specific adaptation; and (3) the full system consistently achieves the highest task success and physical plausibility.

Limitations and Future Work

Our framework also reveals opportunities for future work. Performance is chiefly constrained by the downstream MotionLLM, whose coarse language-to-motion mapping limits fine control—an issue of current executors rather than our reflective planning. While visually grounded feedback avoids the rigidity of geometry-conditioned pipelines, it still lags in geometric precision, and existing contact-focused benchmarks underrepresent semantic correctness and reflective reasoning. Moreover, current metrics are not designed for iterative agents, overlooking improvement across rounds and adaptive stopping. Memory growth across iterations remains modest in practice but motivates exploring more structured memory. These limitations highlight promising directions for stronger motion executors, better evaluation protocols, and more capable reflective planners.

5 Conclusion

Although our reflection-based framework demonstrates strong generalization and interpretable long-horizon behavior, several limitations remain. First, the overall performance is fundamentally bounded by the capacity of the downstream MotionLLM. While the LLM exhibits reliable reasoning and step-wise reflection, the motion generator lacks sensitivity to subtle linguistic variations—different motion intents may be quantized into similar token sequences—forming a bottleneck for fine-grained control. Future work will investigate coupling our framework with higher-capacity motion models or continuous, non-quantized motion representations.

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A Metrics Details

We adopt Physical Validity (PV) to quantify body–scene penetration following prior work (Zhao et al., 2023). Let Ψ_O denote the object signed distance field (SDF), $(\cdot)_-$ clip positive distance values to zero, and V be the set of body vertices. The penetration score at a frame is:

$$s_{\text{inter-pene}} = \sum_{v_i \in V} \left| (\Psi_O(v_i))_- \right| . \tag{1}$$

In our setting, the number of frames varies significantly across tasks, and long-horizon sequences often contain large portions of non-interactive movement (e.g., walking or navigation before actual interaction). Directly averaging $s_{\text{inter-pene}}$ over all frames would dilute meaningful penetration events. We therefore report two robust statistics: (1) the **maximum penetration** pe_{max} across all frames, and (2) the **top-10% mean** penetration $\text{pe}_{10\%}$ computed over the highest-penetration frames. These statistics capture the severity and persistence of penetration for long-horizon interaction tasks with highly imbalanced interactive durations.

B Qualitative Examples

B.1 Online Observation Example

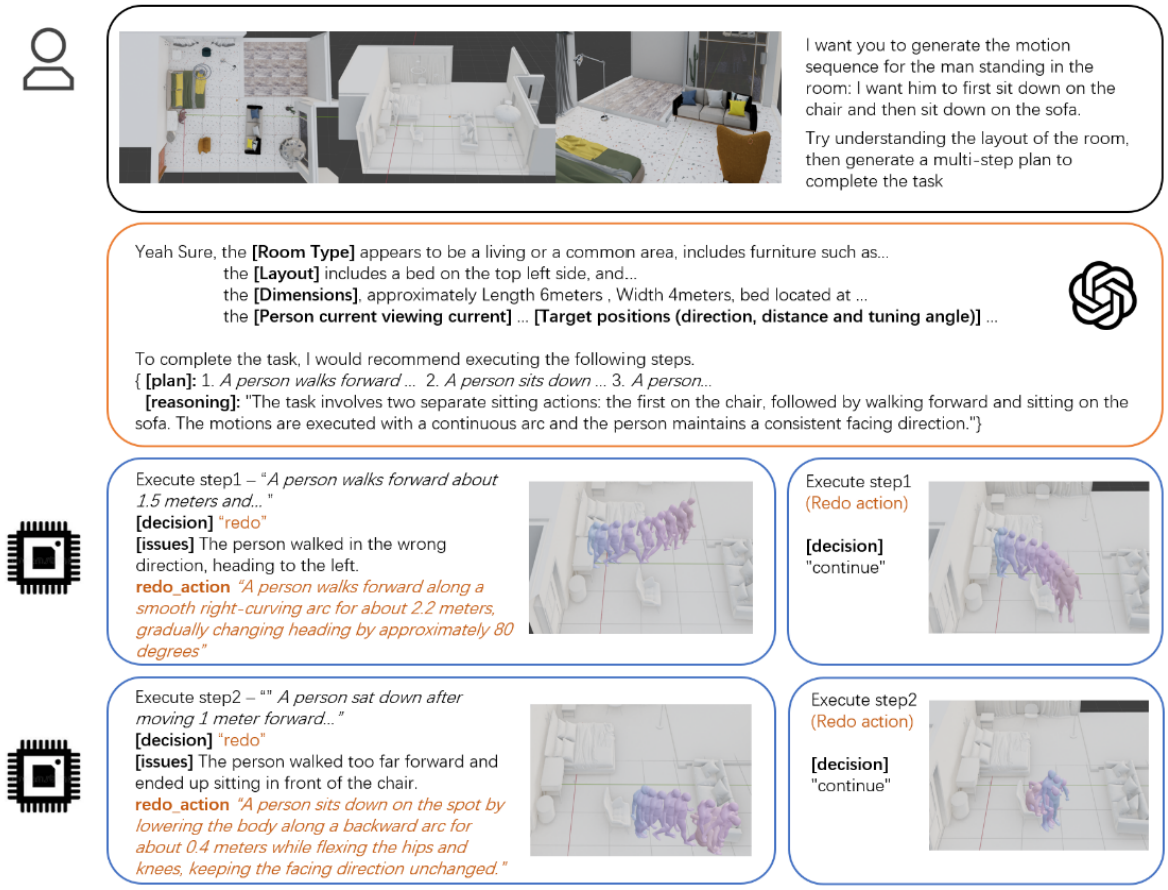

Figure B.1: Online Observation example (steps 1–2).

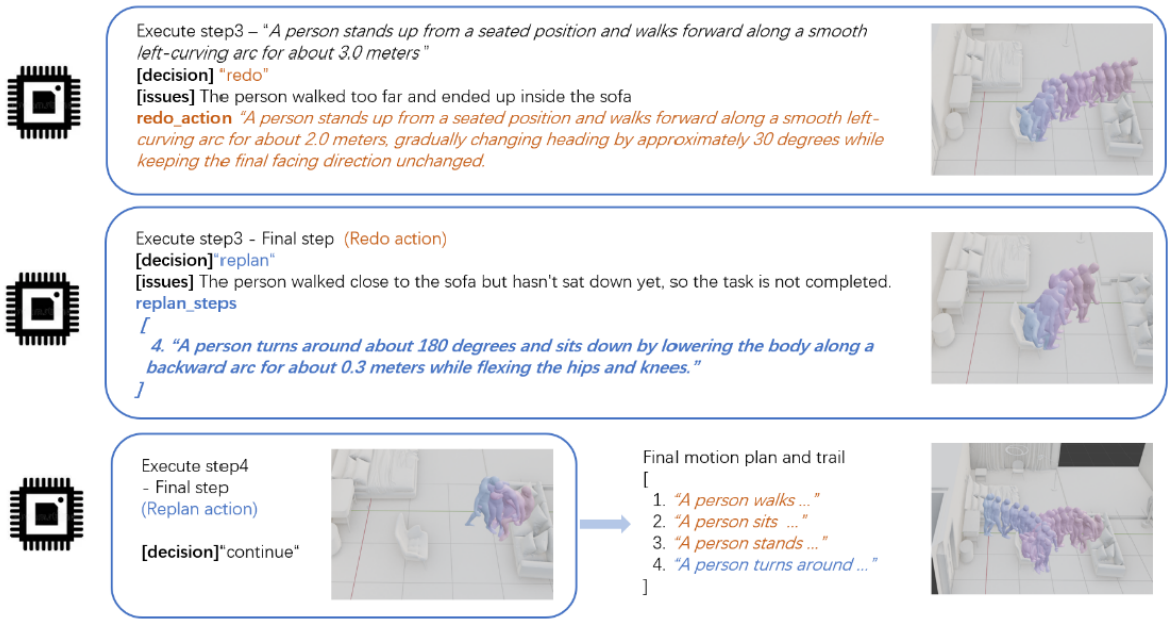

Figure B.2: Online Observation example (steps 3–4 and final plan).

C Prompt Templates

This appendix provides the exact prompts used in our system for (i) scene analysis and planning, (ii) step-wise online observation, and (iii) memory-augmented reflection with adaptive stopping. All placeholders in braces (e.g., `{task_description}`) are replaced by the corresponding runtime content.

C.1 Scene Analysis Prompt (Template 1)

Prompt Template 1 (used).

```
Please analyze these two images: the first is a bird's-eye view,
the second is a first-person view.

Please provide the following information:

1. Room type and layout
2. Room dimensions (length x width, in meters)
3. Current viewing direction (facing where)
4. Distance from current position to various furniture (in meters)
5. Required turning angle to reach target position

Important: The first-person view shows your current facing direction,
which is the starting point for motion planning. Please accurately
determine the current orientation and calculate the required turning
angle to reach the target position.

Note: In first-person view, what you see ahead is your forward direction
(0 degrees), left is -90 degrees, right is +90 degrees. Please calculate
turning angles based on this coordinate system.

Please analyze this scene image and provide a detailed description including:

1. Spatial layout: Room type and overall layout
2. Key objects: Key objects and furniture in the scene
3. Dimensions: Room dimensions and object sizes (in meters)
4. Walkable areas: Walkable areas and paths
5. Obstacles: Obstacles and their positions
6. Height info: Height information for objects and surfaces
7. Starting position: Suggest a reasonable starting position for the person
   (e.g., "entrance", "center of room", "near the door")
8. Spatial metrics for planning:
   - Approximate distances between key objects (in meters)
   - Clearance widths of narrow passages (in meters)
```

```
   - Relative angular directions to target objects (degrees)
   - Whether movement requires sidestepping, turning in place, or wall-following
9. Navigational affordances:
   - Which paths are straight-walkable
   - Which paths require body rotation or lateral movement
   - Which areas are not passable by normal walking

Please provide a comprehensive analysis that can be used for motion planning
and trajectory generation.
```

C.2 Coordinate Trajectory Prompt (Template 2)

Prompt Template 2 (not used in main experiments).

```
You are a trajectory planning expert that helps users generate precise
3D trajectory coordinates for human motion tasks. You will generate a
sequence of 3D coordinate points that represent the person's path.

Task: {task_description}

Scene:

Scene Analysis: {scene_description}

Instructions:
1. Generate a sequence of 3D coordinate points representing the person's trajectory
2. Each point should have x, y, z coordinates in meters
3. Start from a reasonable position (e.g., entrance at x=0, y=0, z=0)
4. Consider the spatial relationships from the scene analysis
5. Use realistic distances and smooth path transitions
6. Include enough points to create a smooth motion (typically 8-20 points)
7. IMPORTANT: Make the trajectory natural and realistic, not just a straight line!

TRAJECTORY DESIGN PRINCIPLES:
- Add natural curves and turns (not just straight lines)
- Include brief pauses or direction changes at key points
- Vary walking speed by adjusting point density (closer points = slower movement)
- Consider obstacles and create realistic navigation paths
- Add small lateral movements (x-axis variations) to simulate natural walking

COORDINATE SYSTEM:
- x: left/right (positive = right)
- y: forward/backward (positive = forward)
- z: up/down (keep at 0 for ground level)

Response Format:
You should respond with a JSON array of coordinate points.
Make sure the trajectory is NOT a straight line:

[
  {"x": 0.0, "y": 0.0, "z": 0.0},
  {"x": 0.2, "y": 0.5, "z": 0.0},
  {"x": 0.8, "y": 1.0, "z": 0.0},
  {"x": 1.5, "y": 1.2, "z": 0.0},
  {"x": 2.2, "y": 1.8, "z": 0.0},
  {"x": 2.8, "y": 2.5, "z": 0.0},
  {"x": 3.0, "y": 3.0, "z": 0.0}
]

Example for "walk to the yellow chair" (assuming chair is at x=3, y=3).
Notice how the path curves naturally and includes lateral movement,
not just forward movement.

Generate your trajectory coordinates now. Remember: make it natural and curved,
not a straight line!
```

C.3 Atomic Motion Plan Prompt (Template 3)

Prompt Template 3 (used for action decomposition).

```
You are a motion planning expert that helps users generate precise human
motion sequences for navigation tasks. You have access to MotionLLM, which
can generate simple, atomic 3D human motions based on textual descriptions.

Task: {task_description}

Scene Analysis: {scene_description}

Instructions:
1. Break down the navigation task into simple, atomic motion steps
2. Each motion step should be described in the format: "A person [action]"
3. Focus on navigation-related actions, such as:
   - turning (left / right / slight)
   - walking forward
   - sidestepping (left / right)
   - stopping or aligning body orientation
4. When applicable, each step MUST include:
   - an approximate distance (in meters), or
   - an approximate turning angle (in degrees).
   Use soft quantities such as "approximately", "about", or "around".
5. Ensure the motion sequence is physically realistic and continuous:
   - Do NOT jump directly to the target.
   - Use intermediate steps for turning, alignment, or obstacle avoidance.
6. Explicitly account for spatial constraints described in the scene analysis:
   - narrow passages
   - obstacles
   - wall-following or clearance constraints
7. Each step must be independently executable by MotionLLM
   (i.e., no compound or ambiguous actions).

Response Format:
You should ONLY respond in JSON format, following this template:
{
  "plan": "A numbered list of motion steps, each in 'A person [action]' format",
  "reasoning": "Brief explanation of the motion sequence logic"
}

Example Response:
{
  "plan": "1. A person turns right by approximately 20 degrees;
           2. A person walks forward approximately 1 meter;
           3. A person sidesteps left approximately 0.5 meters while facing forward;
           4. A person walks forward approximately 1.2 meters;
           5. A person slightly turns left to face the target",
  "reasoning": "The path requires navigating a narrow passage between the table and the wall,
                which prevents direct forward walking."
}

Generate your motion plan now.
```

C.4 Online Observation Prompt (Step-wise Evaluator)

Online Observation Evaluator Prompt.

```
You are an evaluator for step-wise execution in a human-scene interaction task.

You are given:
(1) the high-level task goal,
(2) the intended atomic action,
(3) ONE ghost image showing the executed motion.

IMPORTANT: How to read the ghost image:
- The scene background is static.
- The human appears multiple times in the same image.
- Color encodes time: RED indicates the earliest pose, PURPLE indicates the latest pose.
- The sequence from red -> purple represents the temporal progression of the motion.

Your responsibilities:
1) Determine whether the executed motion matches the intended atomic action.
2) Check physical feasibility:
```

```
   - body-scene penetration,
   - collision with objects,
   - unstable or implausible poses.
3) Check trajectory correctness:
   - wrong direction,
   - under-shoot or over-shoot,
   - accumulated drift relative to the task goal.
4) Decide one of the following:
   - "continue": the action is correct and safe.
   - "redo": this atomic action is incorrect and should be regenerated locally.
   - "replan": this step causes global trajectory drift; keep completed motion,
               but regenerate remaining actions.

You MUST reason based only on the ghost image and the provided text.

Task: {task_description}
Atomic action: {action_pk}

Return JSON only, using the following schema:
{
  "decision": "continue | redo | replan",
  "score": float,   // 0.0-1.0 confidence
  "issues": [
    {
      "type": "collision | penetration | drift | undershoot | overshoot | wrong_direction | unstable_pose",
      "evidence": "brief visual evidence grounded in the ghost image"
    }
  ],
  "redo_action": "A person ...",   // required if decision == "redo", else ""
  "replan_hint": "constraint or correction for replanning",  // required if decision == "replan", else ""
  "stop": false
}
```

C.5 Memory-Augmented Reflection Prompt (Trajectory-level Refinement)

Memory-Augmented Reflection Prompt.

```
You are a motion planning expert that helps users refine navigation motion plans
using feedback from previous execution attempts. You have access to MotionLLM,
which can generate simple, atomic 3D human motions based on textual descriptions.

Task: {task_description}

Scene Analysis: {scene_description}

Trajectory Memory (previous attempts): {trajectory_memory}
Each memory item may include:
- previous plan (steps)
- execution observation (e.g., collision, penetration, stuck, overshoot/undershoot)
- evaluator feedback or failure tags
- final pose/orientation notes (optional)

Instructions:
1. Read the Trajectory Memory and identify what went wrong in the last attempt(s).
   Focus on the most likely failure mode(s), such as:
   - collision with an obstacle
   - penetration into furniture/walls
   - stuck in a narrow passage / insufficient clearance
   - wrong facing direction / misalignment
   - overshoot / undershoot walking distance
   - wrong route choice (approach angle/path selection)
2. Propose a revised strategy that directly addresses the failure(s). Examples:
   - add clearance by inserting a sidestep before entering a narrow passage
   - change the approach angle (turn earlier/later) to avoid cutting corners
   - split a long walk into shorter segments with re-alignment between segments
   - use wall-following to maintain clearance and reduce penetration risk
3. Generate a NEW motion plan (do NOT repeat the old plan verbatim).
4. Break down the navigation task into simple, atomic motion steps.
5. Each motion step should be described in the format: "A person [action]".
6. Focus on navigation-related actions, such as:
```

```
   - turning (left / right / slight)
   - walking forward
   - sidestepping (left / right)
   - stopping or aligning body orientation
7. When applicable, each step MUST include:
   - an approximate distance (in meters), or
   - an approximate turning angle (in degrees).
   Use soft quantities such as "approximately", "about", or "around".
8. Ensure the motion sequence is physically realistic and continuous:
   - Do NOT jump directly to the target.
   - Use intermediate steps for turning, alignment, or obstacle avoidance.
9. Explicitly account for spatial constraints described in the scene analysis:
   - narrow passages
   - obstacles
   - wall-following or clearance constraints
10. Each step must be independently executable by MotionLLM
    (i.e., no compound or ambiguous actions).
11. Keep the revised plan concise (typically 5-12 steps) unless repeated failures indicate
    more fine-grained control is needed.

Response Format:
You should ONLY respond in JSON format, following this template:
{
  "failure_analysis": "A brief diagnosis of what failed in prior attempt(s), grounded in the memory",
  "strategy": "The concrete change(s) you will apply in the new plan",
  "plan": "A numbered list of motion steps, each in 'A person [action]' format",
  "delta": "A short description of how this plan differs from the previous one(s)"
}

Example Response:
{
  "failure_analysis": "The previous attempt clipped into the table corner when entering the passage
                       and ended misaligned, suggesting insufficient clearance and a too-sharp approach angle.",
  "strategy": "Approach the passage with a gentler angle, add a small sidestep to increase clearance,
               and split a long walk into shorter segments with a re-alignment step.",
  "plan": "1. A person turns right by approximately 15 degrees;
           2. A person walks forward approximately 0.8 meters;
           3. A person sidesteps left approximately 0.4 meters while facing forward;
           4. A person walks forward approximately 0.7 meters;
           5. A person stops and slightly turns left by approximately 10 degrees to re-align;
           6. A person walks forward approximately 0.8 meters toward the target;
           7. A person stops and aligns body orientation to face the target",
  "delta": "Added a 0.4m sidestep before the narrow passage, reduced entry turning angle,
            and split a long forward walk into two shorter segments with mid-course re-alignment."
}

Generate your revised motion plan now.
```

C.6 Adaptive Stopping Check Prompt

Stop Check Prompt (Pass/Fail).

```
You are judging whether the task has been successfully completed.

Given:
- the task description,
- the final executed trajectory shown as a ghost image.

IMPORTANT:
- RED = earliest pose, PURPLE = final pose.

Determine:
1) Whether the final (purple) pose satisfies the task goal.
2) Whether any severe penetration or collision occurs along the trajectory.

Return JSON only:
{
  "status": "Pass | Fail",
  "reason": "one-sentence justification based on the ghost image"
}
```